

Development Automation **& Linux Lab**

Experiment 7

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EXPERIMENT-7

Working with Make and Makefile

1. Write a C program for calculator with basic functionality subtraction, addition, division, and multiplication. Each functionality (addition, subtraction, etc.) should be written in separate source files along with a single main source file. Use make utility to create an executable file for the calculator.

Functionality Codes:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7a$ cat add.c
#include<stdio.h>
#include"header.h"
void add(int a, int b){
    printf("%d",a+b);
}
```

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7a$ cat sub.c
#include<stdio.h>
#include"header.h"
void sub(int a, int b){
    printf("%d",a-b);
}
```

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7a$ cat mul.c
#include<stdio.h>
#include"header.h"
void mul(int a, int b){
    printf("%d",a*b);
}
```

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7a$ cat div.c
#include<stdio.h>
#include"header.h"
void div(int a, int b){
    printf("%d",a/b);
}
```

Header file:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7a$ cat header.h
void add(int a, int b);
void sub(int a, int b);
void mul(int a, int b);
void div(int a, int b);
```

Main file:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7a$ cat main.c
#include<stdio.h>
#include"header.h"

int main(){
    int a, b;
    char op;
    printf("Enter 2 Numbers: ");
    scanf("%d %d", &a, &b);
    printf("Enter Operation (+, -, *, /): ");
    scanf(" %c",&op);

    switch (op){
        case '+':
            add(a,b);
            break;
        case '-':
            sub(a,b);
            break;
        case '*':
            mul(a,b);
            break;
        case '/':
            div(a,b);
            break;
        default:
            printf("Invalid Operation.\n");
            return 1;
    }
    printf("\n");
    return 0;
}
```

Make file:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7a$ cat makefile
calculator: main.o add.o sub.o mul.o div.o
    gcc main.o add.o sub.o mul.o div.o -o calculator

clean:
    rm -rf *.o final
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7a$ make calculator
cc -c -o main.o main.c
cc -c -o add.o add.c
cc -c -o sub.o sub.c
cc -c -o mul.o mul.c
cc -c -o div.o div.c
gcc main.o add.o sub.o mul.o div.o -o calculator
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7a$ ./calculator
Enter 2 Numbers: 2 3
Enter Operation (+, -, *, /): *
6
```

2. In the above program include two additional features for computing power and modulus of given numbers in the calculator program. Use Make utility to automatically build the second version of the calculator with additional features included in it.

New Functionality Codes:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7b$ cat mod.c
#include<stdio.h>
#include"header.h"
void mod(int a, int b){
    printf("%d",a%b);
}
```

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7b$ cat powe.c
#include<stdio.h>
#include<math.h>
#include"header.h"

void powe(int a, int b){
    int x=a;
    for(int i=1;i<b;i++){
        x=x*a;
    }
    printf("%d", x);
}
```

Updated Header file:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7b$ cat header.h
void add(int a, int b);
void sub(int a, int b);
void mul(int a, int b);
void div(int a, int b);
void mod(int a, int b);
void powe(int a, int b);
```

Updated Main file:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7b$ cat main.c
#include<stdio.h>
#include"header.h"

int main(){
    int a, b;
    char op;
    printf("Enter 2 Numbers: ");
    scanf("%d %d", &a, &b);
    printf("Enter Operation (+,-,*,/,^,%): ");
    scanf(" %c",&op);

    switch (op){
        case '+':
            add(a,b);
            break;
        case '-':
            sub(a,b);
            break;
        case '*':
            mul(a,b);
            break;
        case '/':
            div(a,b);
            break;
        case '^':
            powe(a,b);
            break;
        case '%':
            mod(a,b);
            break;
        default:
            printf("Invalid Operation.\n");
            return 1;
    }
    printf("\n");
    return 0;
}
```

Updated Make file:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7b$ cat makefile
calculator: main.o add.o sub.o mul.o div.o
    gcc main.o add.o sub.o mul.o div.o -o calculator

calculator2: main.o add.o sub.o mul.o div.o mod.o powe.o
    gcc main.o add.o sub.o mul.o div.o mod.o powe.o -o calculator2

clean:
    rm -rf *.o final
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7b$ make calculator2
cc -c -o main.o main.c
cc -c -o add.o add.c
cc -c -o sub.o sub.c
cc -c -o mul.o mul.c
cc -c -o div.o div.c
cc -c -o mod.o mod.c
cc -c -o powe.o powe.c
gcc main.o add.o sub.o mul.o div.o mod.o powe.o -o calculator2
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7b$ ./calculator2
Enter 2 Numbers: 3 4
Enter Operation (+,-,*,/,^,%): ^
81
```

3. Write Makefile in above program using the best practices.

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7b$ cat makefile
.PHONY = clean
CC = gcc
OBJECTS1 = main.o add.o sub.o mul.o div.o
OBJECTS2 = main.o add.o sub.o mul.o div.o mod.o powe.o

calculator: $(OBJECTS1)
    $(CC) -o calculator $(OBJECTS1)

calculator2: $(OBJECTS2)
    $(CC) -o calculator2 $(OBJECTS2)

clean:
    rm -rf *.o final
```

4. Use make utility to archive the logs file and move it to some other location.

Code:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7/ques4$ cat makefile
LOGS = /var/log/*.log
ARCHIVE = /mnt/c/college/upes/sem4/linux/lab/scripts/exp7/ques4/
FILENAME = archive-$(shell date +%Y%m%d).tar.gz

execute: archive delete

archive:
    tar -czvf $(ARCHIVE)$(FILENAME) $(LOGS)
    echo "Archive Successful!"

delete:
    rm -f $(LOGS)
    echo "Logs Deleted!"
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7/ques4$ make execute
tar -czvf /mnt/c/college/upes/sem4/linux/lab/scripts/exp7/ques4/archive-20230322.tar.gz /var/log/*.log
tar: Removing leading '/' from member names
/var/log/alternatives.log
tar: Removing leading '/' from hard link targets
/var/log/bootstrap.log
/var/log/dpkg.log
/var/log/fontconfig.log
/var/log/ubuntu-advantage.log
/var/log/upgrade-policy-changed.log
echo "Archive Successful!"
Archive Successful!
rm -f /var/log/*.log
rm: cannot remove '/var/log/alternatives.log': Permission denied
rm: cannot remove '/var/log/bootstrap.log': Permission denied
rm: cannot remove '/var/log/dpkg.log': Permission denied
rm: cannot remove '/var/log/fontconfig.log': Permission denied
rm: cannot remove '/var/log/ubuntu-advantage.log': Permission denied
rm: cannot remove '/var/log/upgrade-policy-changed.log': Permission denied
make: *** [makefile:12: delete] Error 1
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7/ques4$ ls
archive-20230322.tar.gz makefile
```

5. Demonstrate the usage of conditional statements in a Makefile for a simple C/shell script program.

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7/ques5$ cat makefile
all: hello

ifdef DEBUG
CFLAGS=-g
endif

hello: hello.c
    gcc $(CFLAGS) -o hello hello.c

clean:
    rm -f hello
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7/ques5$ make
gcc -o hello hello.c
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp7/ques5$ ./hello
Hello, world!
```